B.N.M Institute of Technology

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Vidyayāmruthamashnuthe

INTERACTIVE HEALTH BOT

By

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Project Guide

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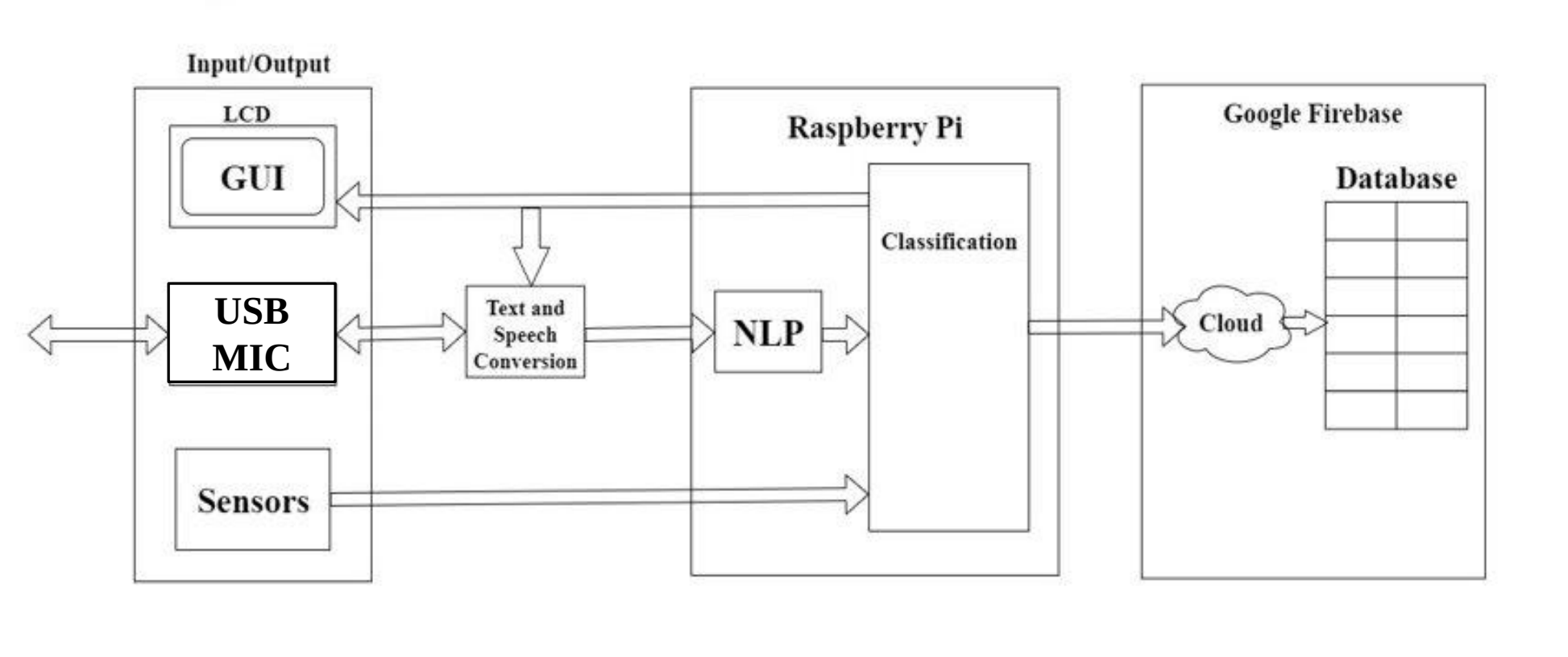
PROJECT OBJECTIVES

- Build a Raspberry pi based health monitor with sensors for capturing vital signs
- Implement a device that intelligently guides the user through a voice based interaction
- Build a medical corpus of diagnostic information connecting symptoms with diseases
- Set up and use a backend as a service to store patient diagnostics history
- Build a web based application to pull patient data from backend as a service

BLOCK DIAGRAM



INTRODUCTION

OVERVIEW OF THE PROJECT FLOW (PHASE 1)

- Literature survey
- Formulation of the Design (Hardware and Software)
- Proposed methods
- Compiled list of diseases (Initially only 23 diseases)
- Obtained corresponding dataset
- Started with the implementation of NLP (Natural Language Processing)

INTRODUCTION

OVERVIEW OF THE PROJECT FLOW (PHASE 2)

- Compiled final list of **42 diseases** and its corresponding data set
- Implemented **NLP (Natural Language Processing)** and tested for its accuracy
- Incorporated **Classification** algorithm
- Developed **GUI (Graphical User Interface)**
- Rigorous **testing** of the complete code (NLP + Classification + GUI).
- Obtained convincing **results**
- **Temperature sensor** testing done separately with Raspberry PI and obtained accurate results.

INTRODUCTION

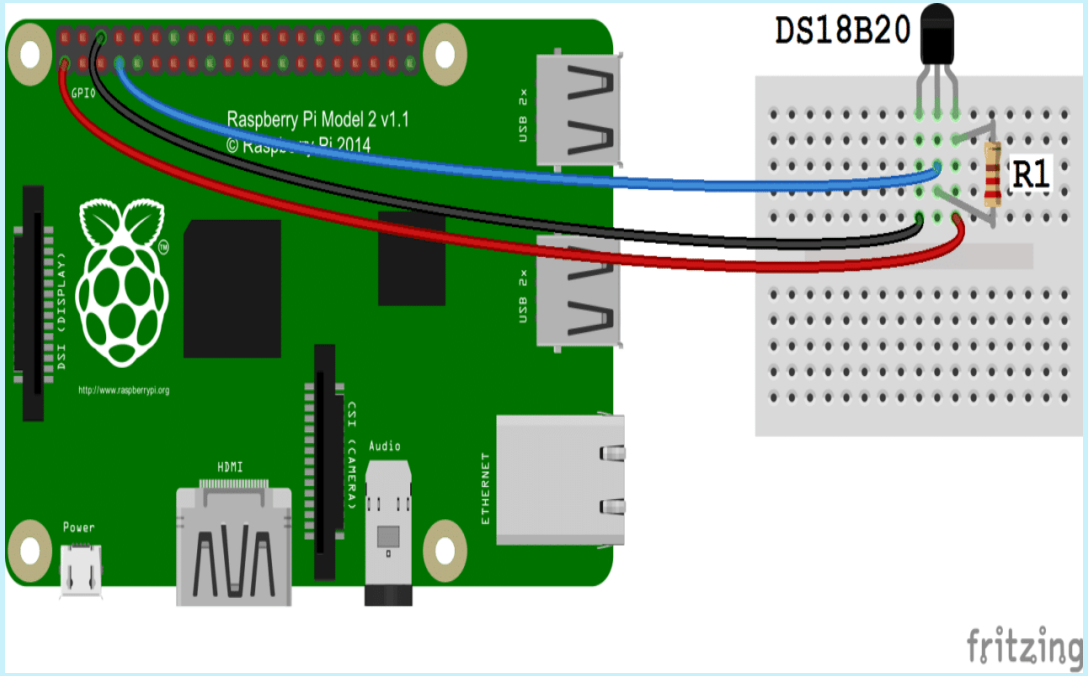
OVERVIEW OF THE PROJECT FLOW (FINAL)

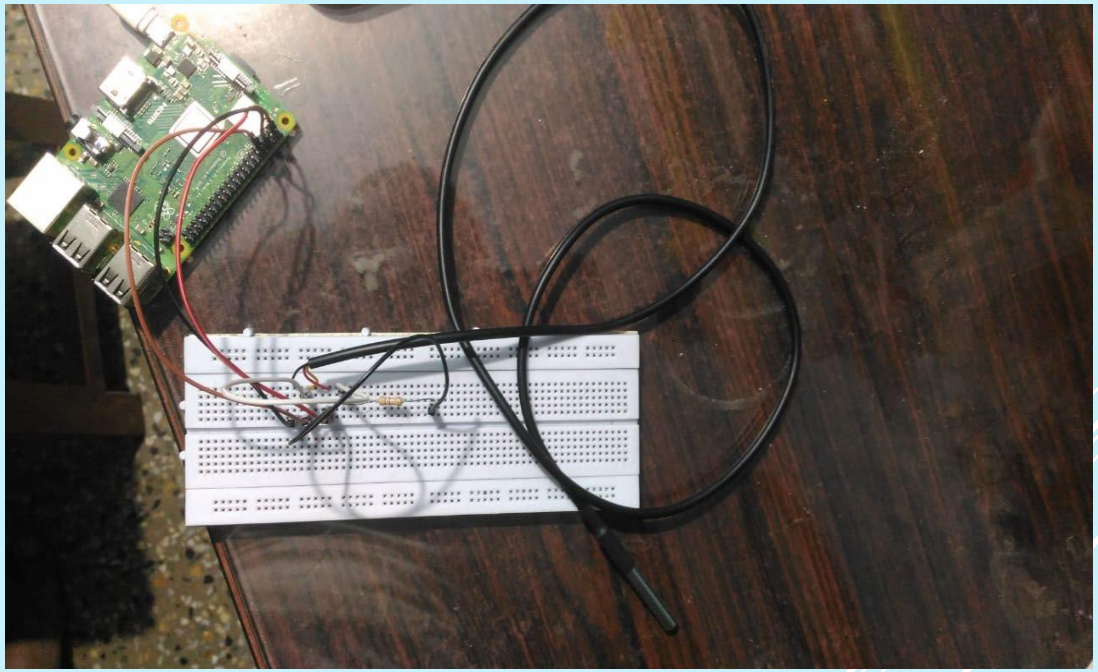
- Improvised the code to remove the loop holes found earlier.
- Improvised the **GUI appearance** to make it more appealing.
- Implemented **"Special / Unique" Questions** to enhance the detection capabilities.
- NLP corrections done to recognize the symptoms more effectively.
- Added **Firebase** feature for further data analytics.
- Completion of **Hardware Setup** including the Display and the USB MIC.
- Complete Hardware and Software code **testing**.
- **Obtained convincing results** and the **accuracy is improved** with the help of **special questions feature.**

HARDWARE

- Raspberry Pi 3 Model B+
- DS18B20 Waterproof Temperature sensor probe
- 5 inch HDMI Raspberry Pi Display
- USB Microphone
- Speakers

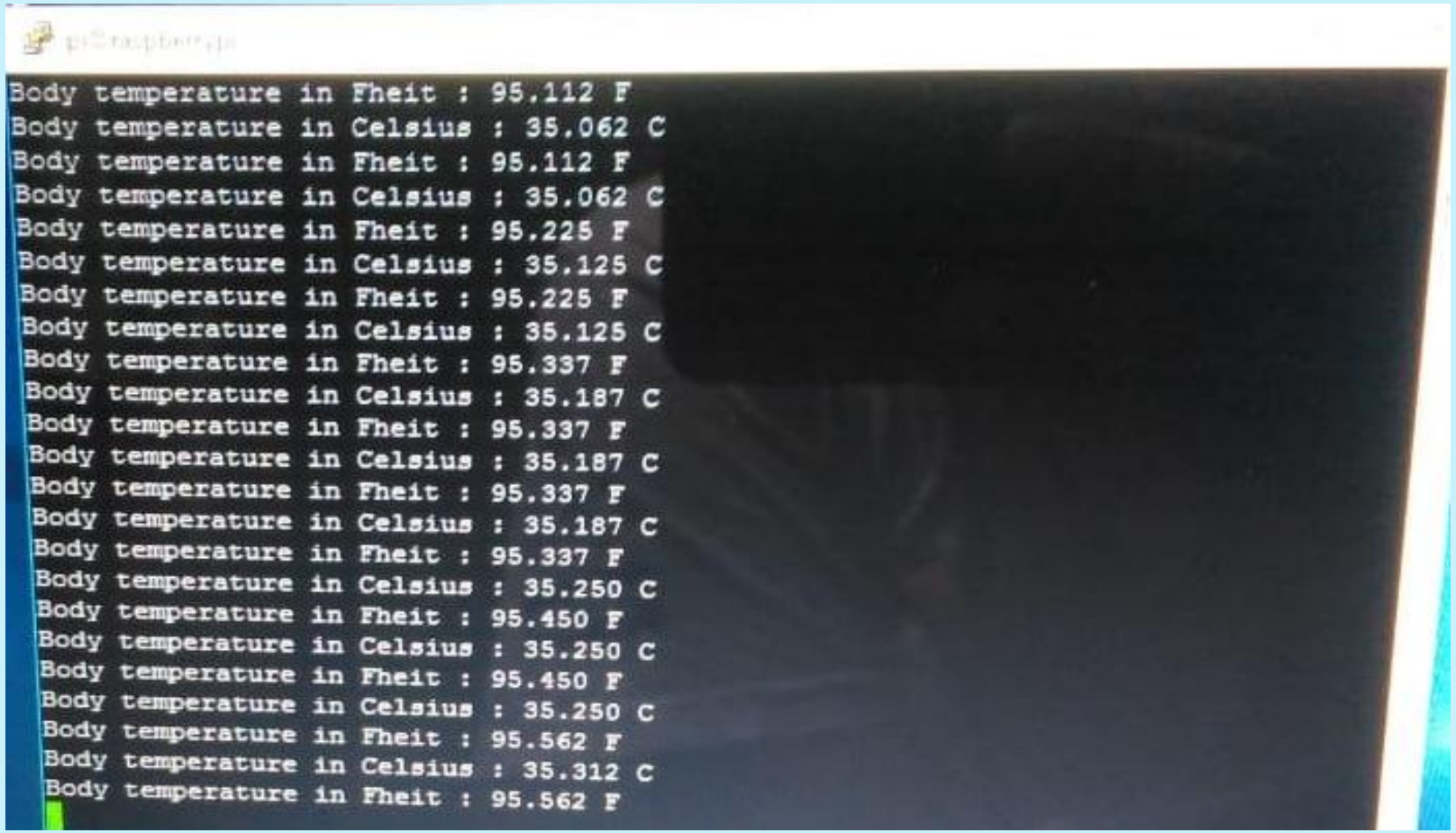
DS18B20 WATERPROOF TEMPERATURE SENSOR





2D and 3D view of the DS18B20 Temperature Sensor circuit interface with RPi

DS18B20 WATERPROOF TEMPERATURE SENSOR



Results obtained from DS18B20 Waterproof Temperature Sensor

OTHER COMPONENTS

USB MIC

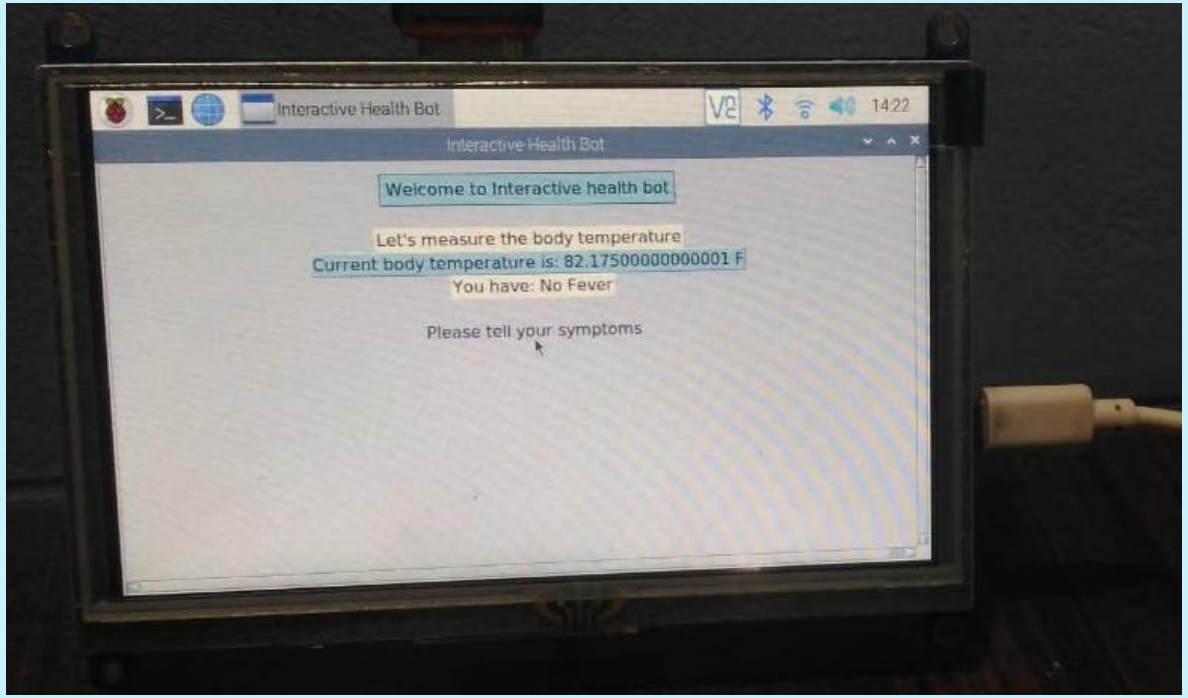
1. 5 inch HDMI Raspberry Pi Display

- 5 inch wide display with HDMI support.
- Simple interface with Raspberry Pi
- HDMI support enables the usage of GPIO pins for the sensor.
- Touch screen option available but not used in this project.

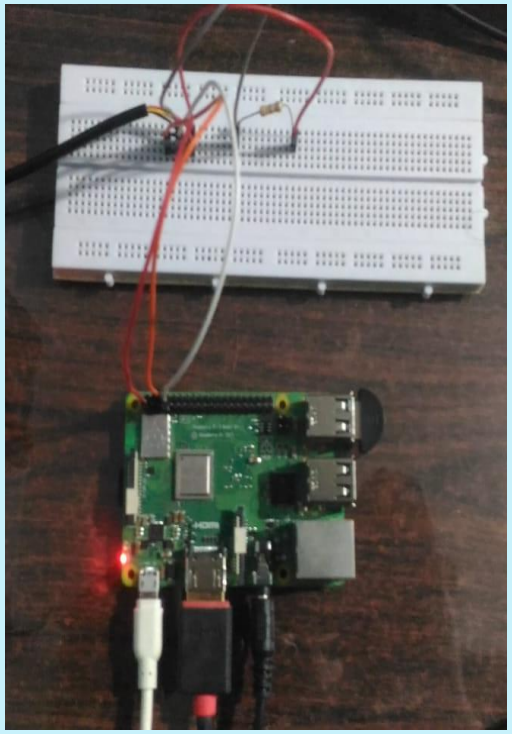
2. USB Microphone (MIC)

- Small, miniature USB device for Audio Input to the Raspberry Pi
- Quality of the audio captured is good.

OTHER COMPONENTS



5 inch LCD display



Temp sensor circuit

COMPLETE SETUP

Complete setup

USER INTERFACE - NLP

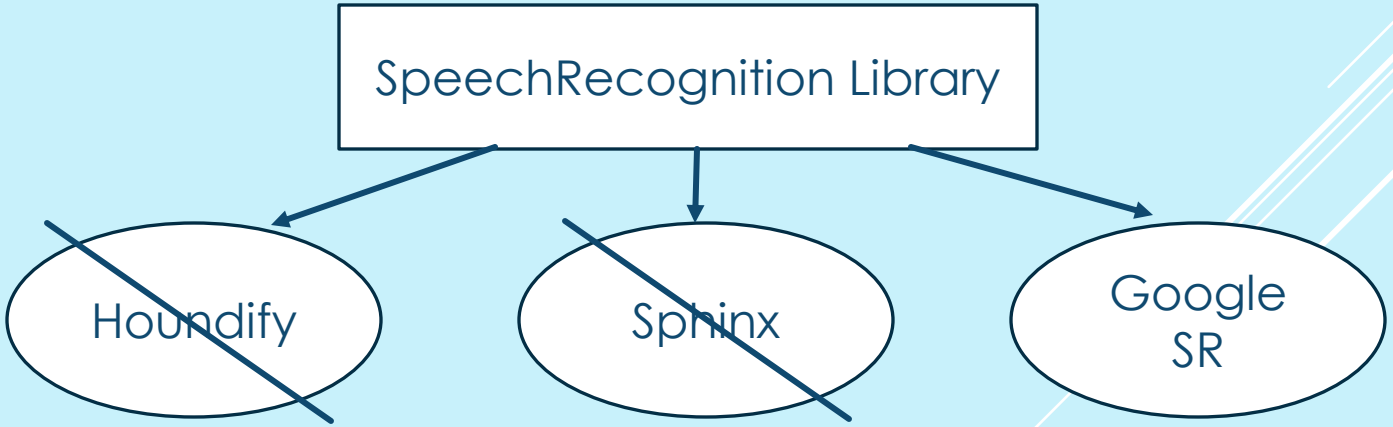
Steps:

1. Speech Synthesis & Recognition
2. Modifying the dataset
3. Interpreting the input
4. Review

NLTK Package

1. Speech Synthesis and Recognition

Speech to text:



Text to Speech:

- pyttsx3

2. Modifying the dataset → Stemming using PorterStemmer

3. Interpreting the input

Tokenization
↓
Stop words removal
↓
PoS tagging
↓
Getting the synonyms
↓
Stemming
↓
N - grams
↓
Random shuffle

Input - "I am facing severe pain in the chest and I am short of breath"

↓

[“I” , “ am” , “facing” , “severe” , “pain”, “in” , “the” , “chest” , "and" , "I" , "am" , "short" , "of" , "breath"]

↓

[“facing” , “severe” , “pain”, “chest” , "short" , "breath"]

↓

["VBG" , "JJ" , "NN" , “NN" , "JJ" , "NN"]

STRING MATCHING

- To obtain close matches
- **Difflib**
  - Computes difference between 2 sequences
  - Depends on the order

E.g:

A = " watering of eyes"

B = "watery eyes"

Result = **"watering of eyes"**

B = "eyes are watery"

Result = **No match**

- **Fuzz module**
  - Uses Levenshtein distance
  - Threshold = score out of 100
  - Fuzz partial ratio

E.g:

A = " watering of eyes"

B = "watery eyes"

Threshold = **38**

Result = **"watering of eyes"**

Threshold = **80**

Result = **No match**

PHONETIC MATCHING ALGORITHMS USED

Soundex

- 4 character long alphanumeric string that represents English pronunciation of a word
- Similar sounding words share the same key

Metaphone

- Improves on Soundex – generates variable length codes
- More accurate

Words	soundex()	metaphone()
Pain Pane	P500 P500	PN PN
Sneeze Freeze	S520 F620	SNS FRS
Muscle paining Muscle wasting	M241 M242	MSKL PNNK MSKL WSTNK

CLASSIFICATION MODELS USED

Decision Tree

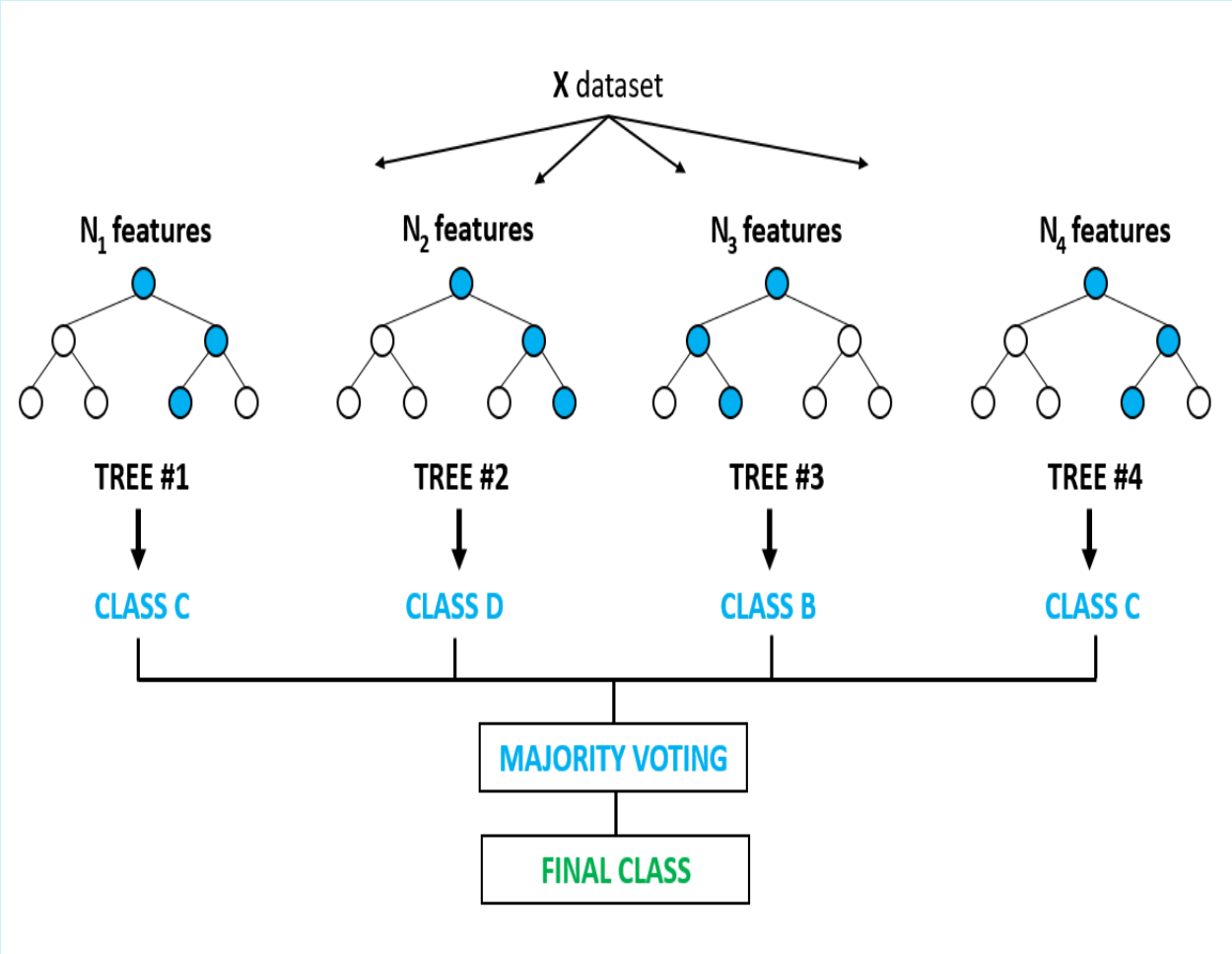
1. Model built.
   Random predictions with incorrect results.

Random Forest Classifier

1. Model built.
   Symptom inputting and checking was long and tedious.
   Model not very user-friendly.

2. Model built.
   Code shows promising result with some minor issues.
   Issues solved after modification.
   User-friendly model.
   Accuracy currently at 95%

RANDOM FOREST CLASSIFIER



- A random forest is a meta estimator that fits a number of decision tree classifiers on various sub-samples of the dataset and uses averaging to improve the predictive accuracy and control over-fitting.
- Random sampling of training data points when building tree.
- Random subsets of features considered when splitting nodes

RANDOM FOREST CLASSIFIER

- The idea is that by training each tree on different samples, although each tree might have high variance with respect to a particular set of the training data, overall, the entire forest will have lower variance but not at the cost of increasing the bias.
- The procedure of training each individual learner on different bootstrapped subsets of the data and then averaging the predictions is known as "**bagging**" , short for bootstrap aggregating.
- At test time, predictions are made by averaging/majority the predictions of each decision tree.

RANDOM FOREST CLASSIFIER

- **n_estimators**: The number of trees in the forest. The number used by us is **10**.
- **LabelEncoder**:
  - From sklearn for pre-processing.
  - Encode target labels with value between 0 and n_classes-1.
  - This transformer is used to encode target values.

- We are using **RandomForestClassifier** for the its following methods:
  - **self.fit** : Build a forest of trees from the training set X & Y. (X=df, Y=prognosis)
  - **self.predict_proba** : Predict class probabilities for Y= prognosis.

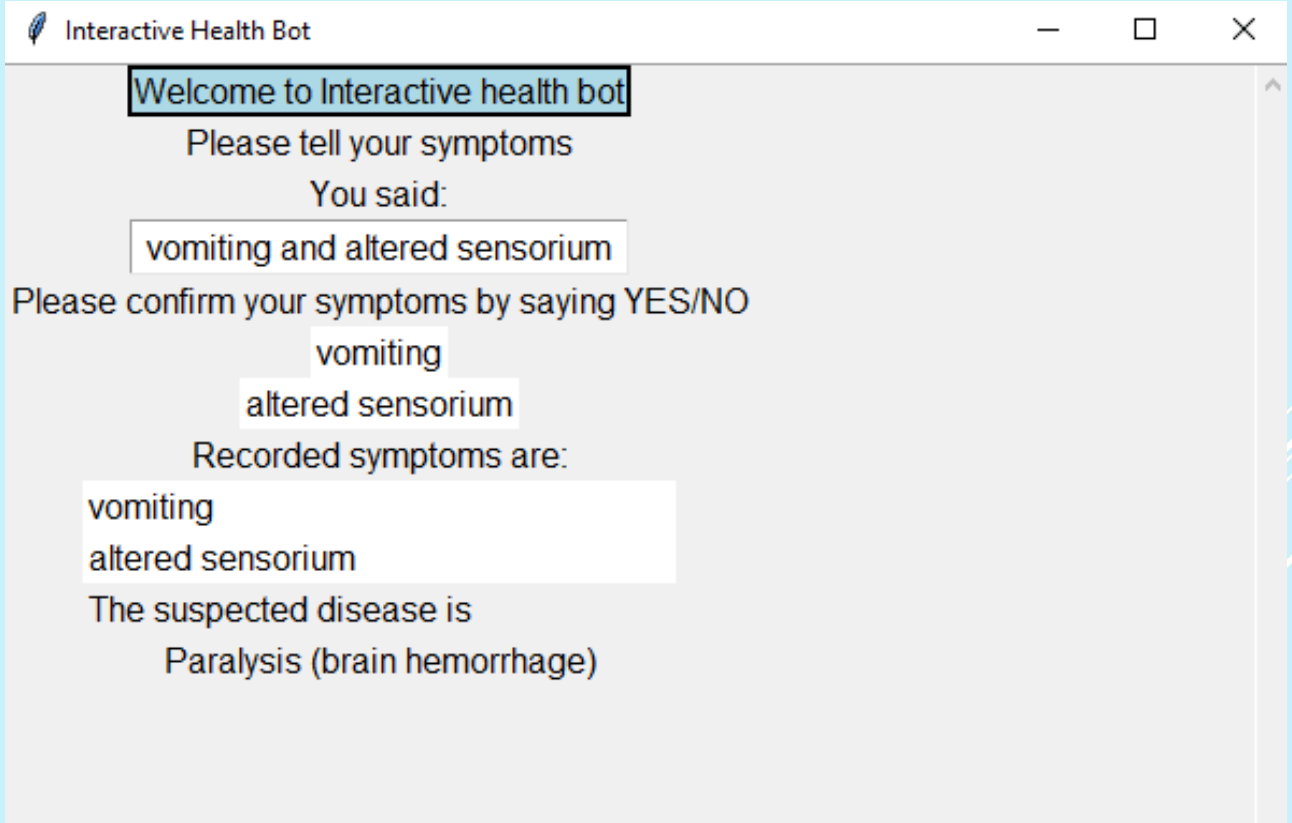
RANDOM FOREST CLASSIFIER

- Information displayed at each iteration:
  - Symptoms covered
  - Diseases probabilities
  - Possible Diseases
  - Remaining Symptoms

- If probability > **0.7** within 3 iterations , the prognosis name is displayed.
- Else, iterations continue for 3 times. Then, if iterations **>3** & probability <**0.7** , prognosis is determined based on special questions.

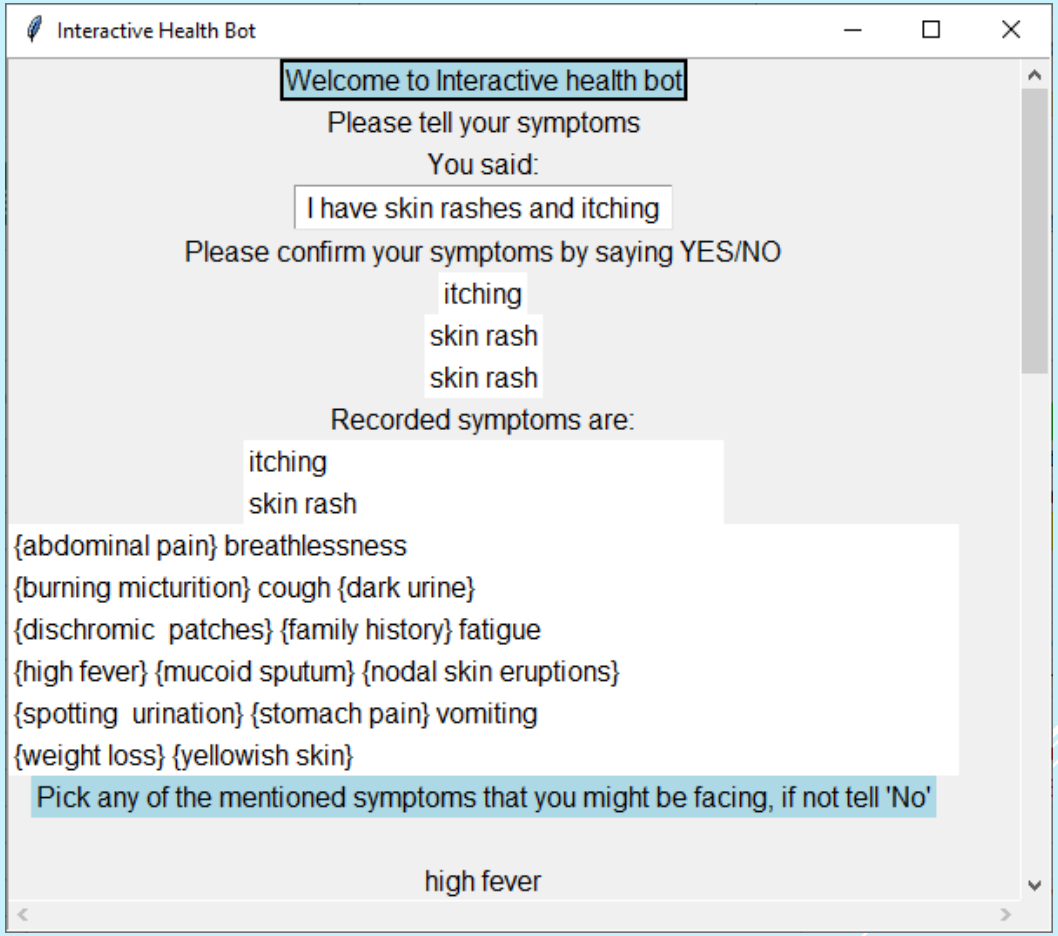
RESULTS

- The results of the project as of now is displayed in the following slides
- Case 1: Disease is predicted only with the initial symptoms



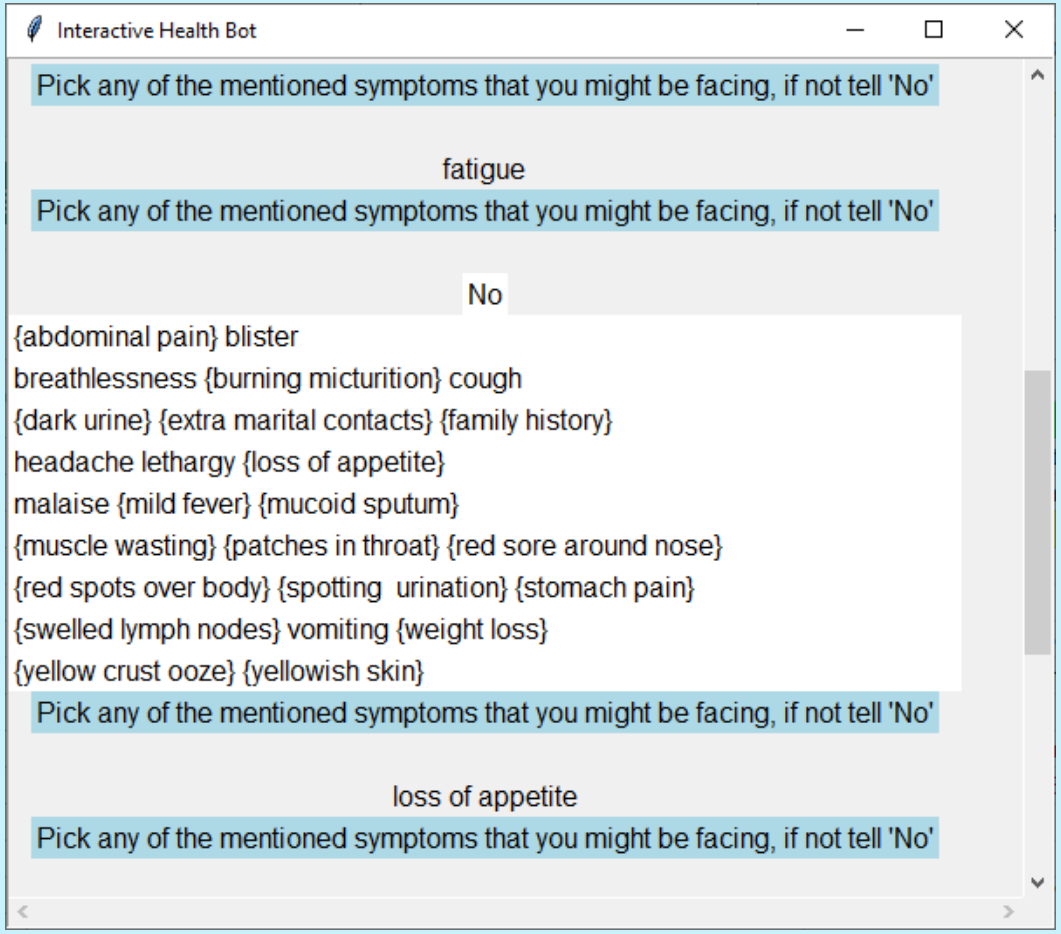
RESULTS

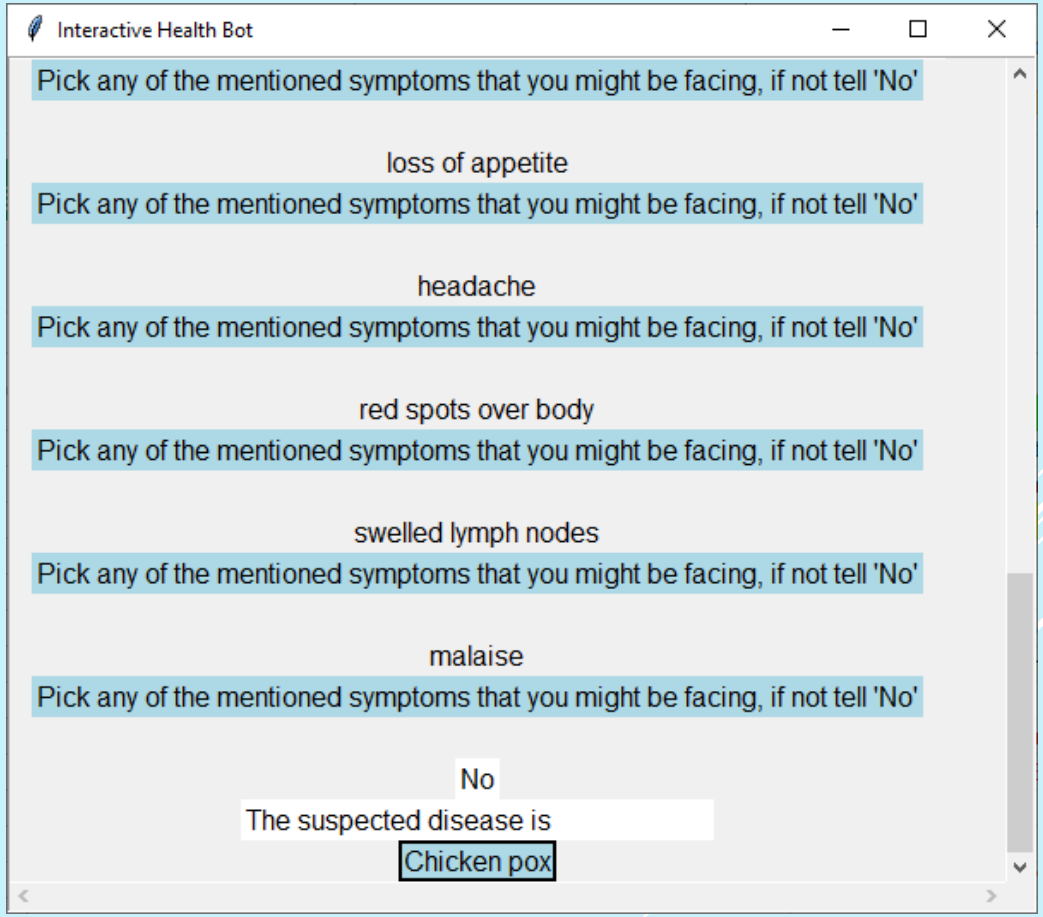
- Case 2: Disease is predicted after few rounds of asking for more symptoms



RESULTS

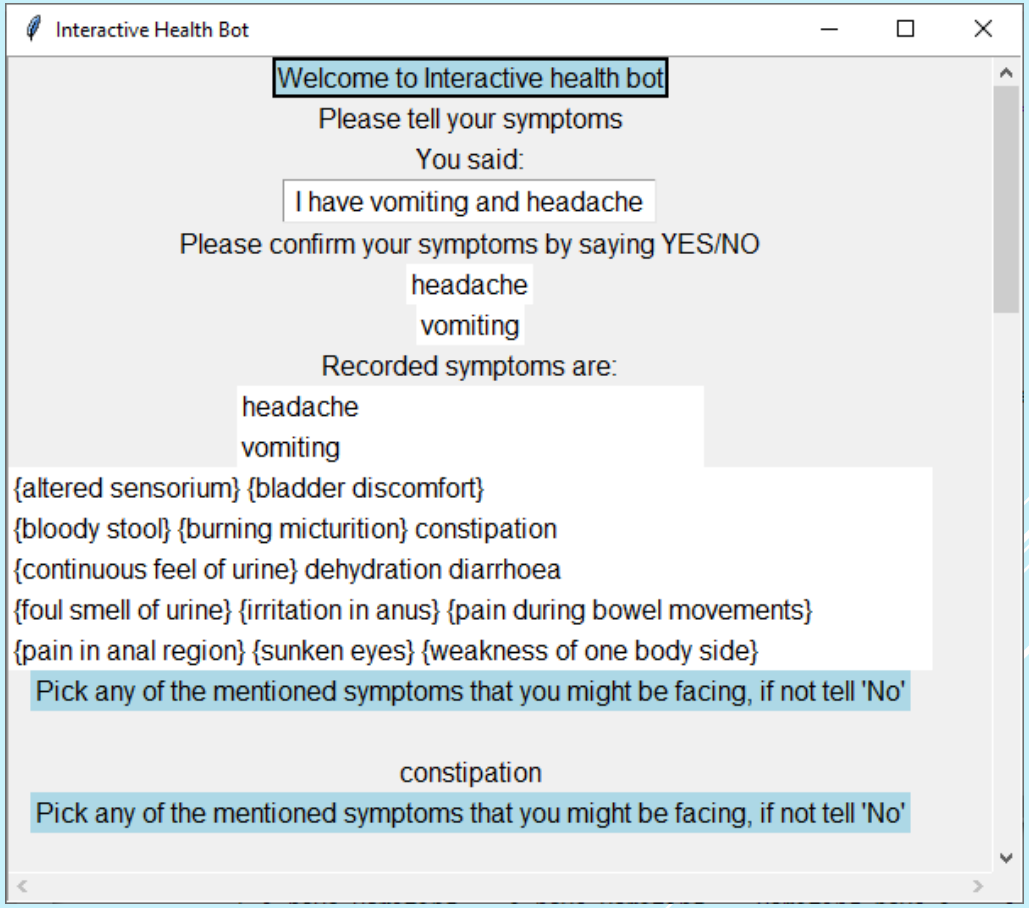
Case 2:





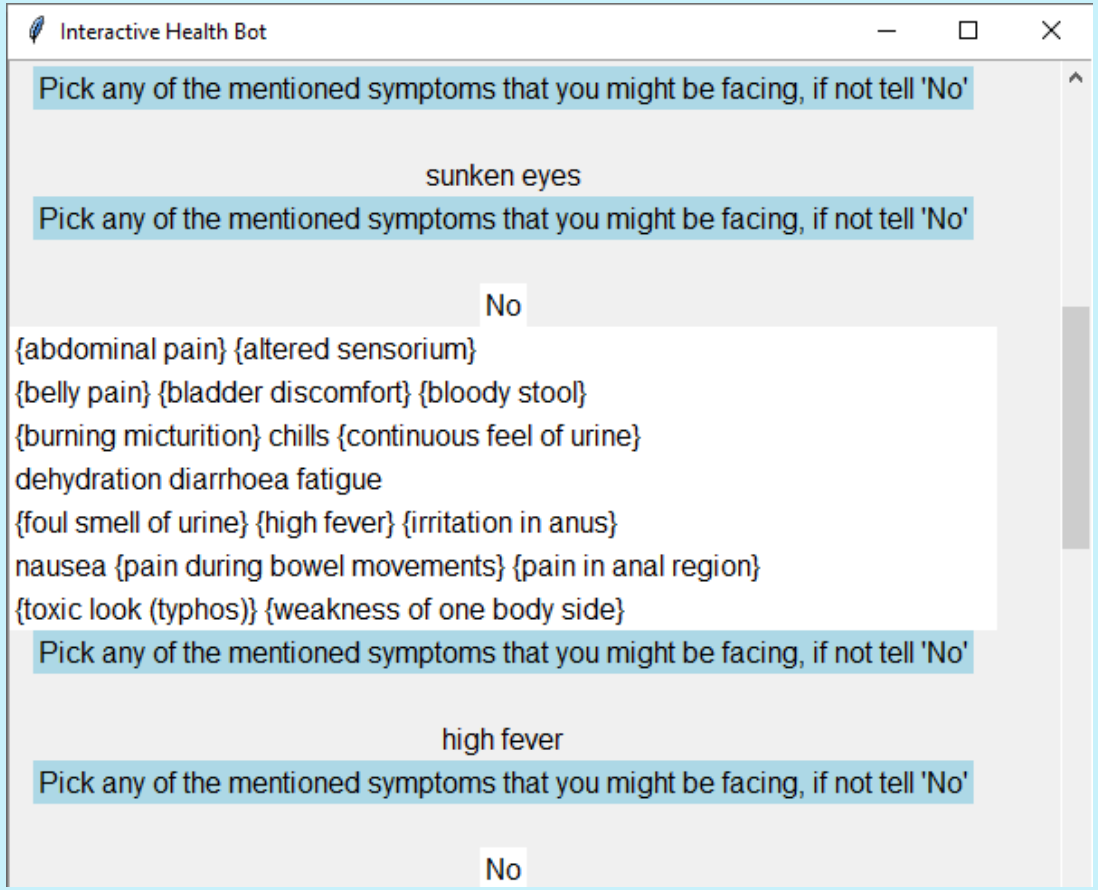
RESULTS

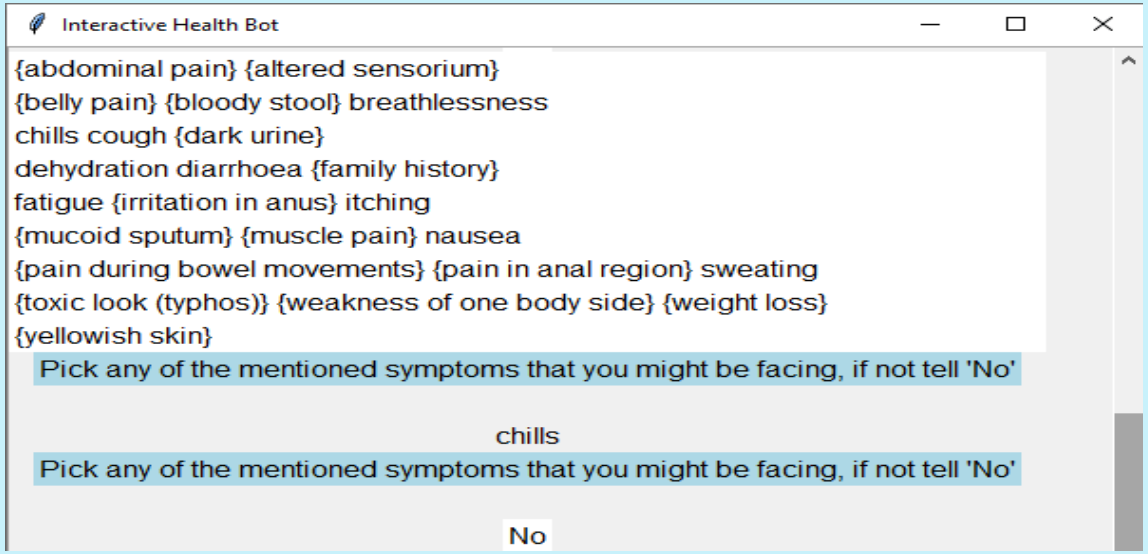
- Case 3: At the end of the trial, a list of possible diseases is given instead of one disease prediction

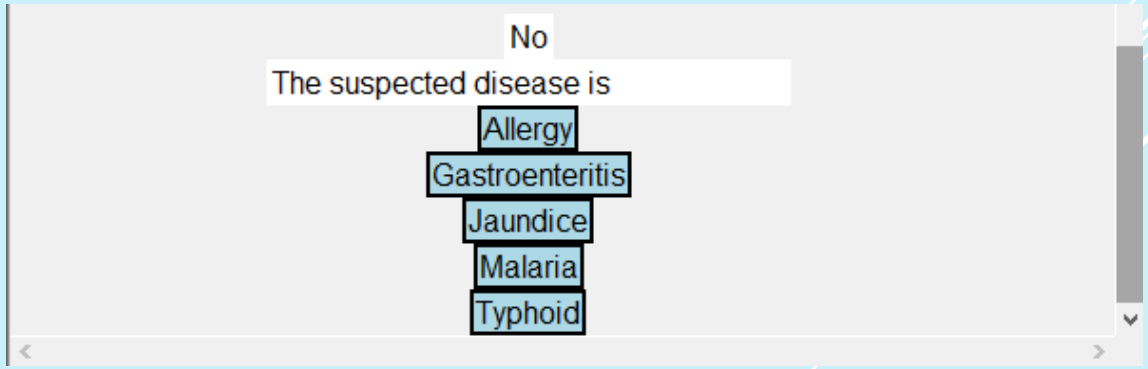


RESULTS

Case 3:

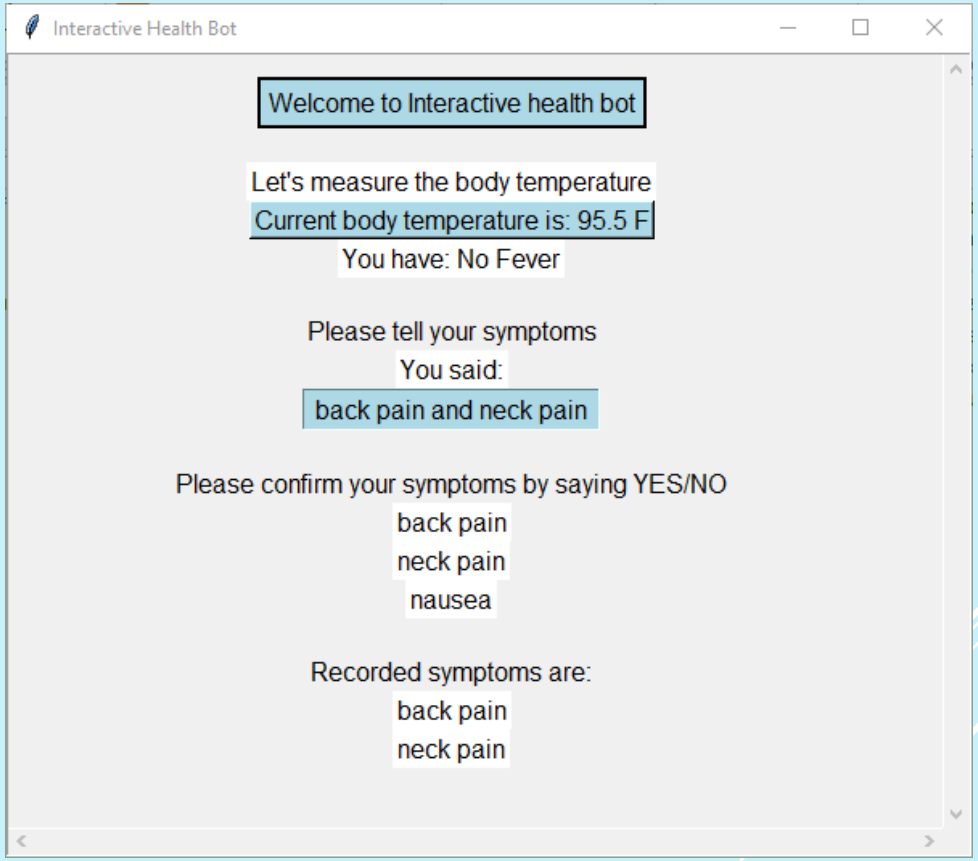




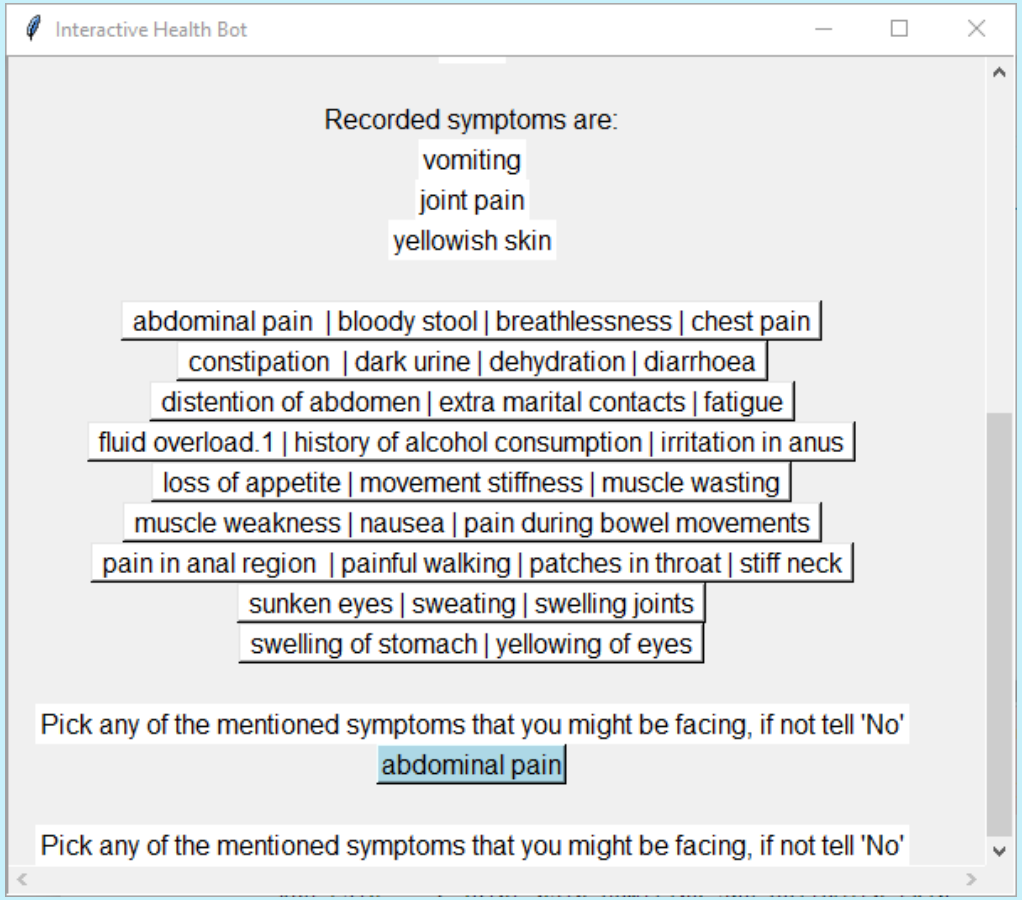


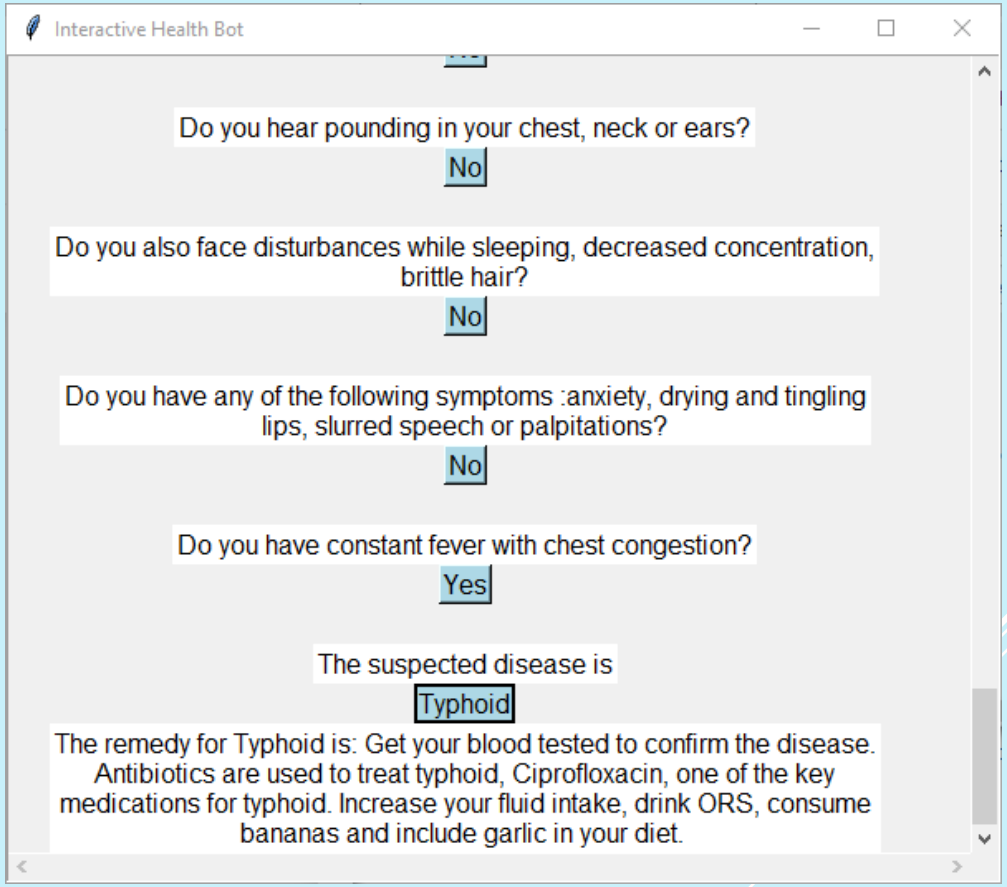
RESULTS

- Current appearance of the GUI window after modification
- Displaying remaining symptoms has been modified
- Special questions are implemented which help in getting better results
- Remedies are displayed for all the diseases



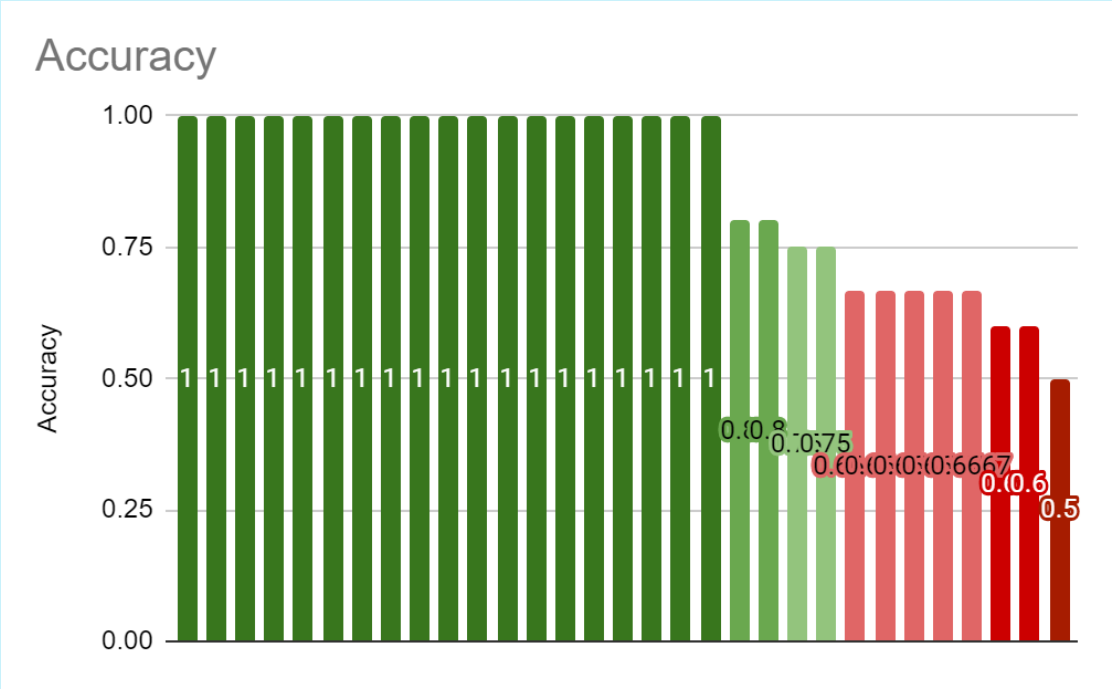
RESULTS


Interactive Health Bot
Recorded symptoms are:
vomiting
joint pain
yellowish skin
abdominal pain | bloody stool | breathlessness | chest pain
constipation | dark urine | dehydration | diarrhoea
distention of abdomen | extra marital contacts | fatigue
fluid overload.1 | history of alcohol consumption | irritation in anus
loss of appetite | movement stiffness | muscle wasting
muscle weakness | nausea | pain during bowel movements
pain in anal region | painful walking | patches in throat | stiff neck
sunken eyes | sweating | swelling joints
swelling of stomach | yellowing of eyes
Pick any of the mentioned symptoms that you might be facing, if not tell 'No'
abdominal pain
Pick any of the mentioned symptoms that you might be facing, if not tell 'No'


Interactive Health Bot
Do you hear pounding in your chest, neck or ears?
No
Do you also face disturbances while sleeping, decreased concentration, brittle hair?
No
Do you have any of the following symptoms :anxiety, drying and tingling lips, slurred speech or palpitations?
No
Do you have constant fever with chest congestion?
Yes
The suspected disease is
Typhoid
The remedy for Typhoid is: Get your blood tested to confirm the disease. Antibiotics are used to treat typhoid, Ciprofloxacin, one of the key medications for typhoid. Increase your fluid intake, drink ORS, consume bananas and include garlic in your diet.

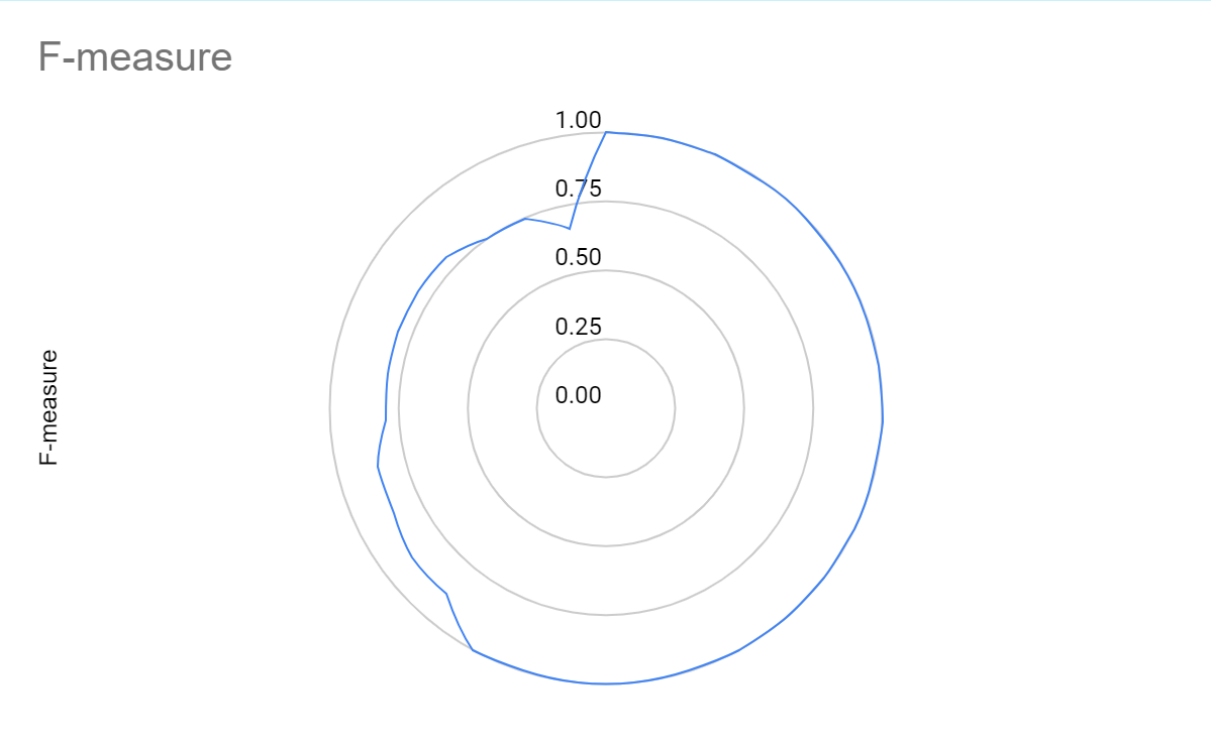
RESULTS

- Accuracy of NLP model:
  - Several tests were conducted
  - Based on the number of symptoms correctly recognized each test was given a value between 0 and 1
  - Average of it was found to be 0.8752 or 87.52% which is considered the accuracy



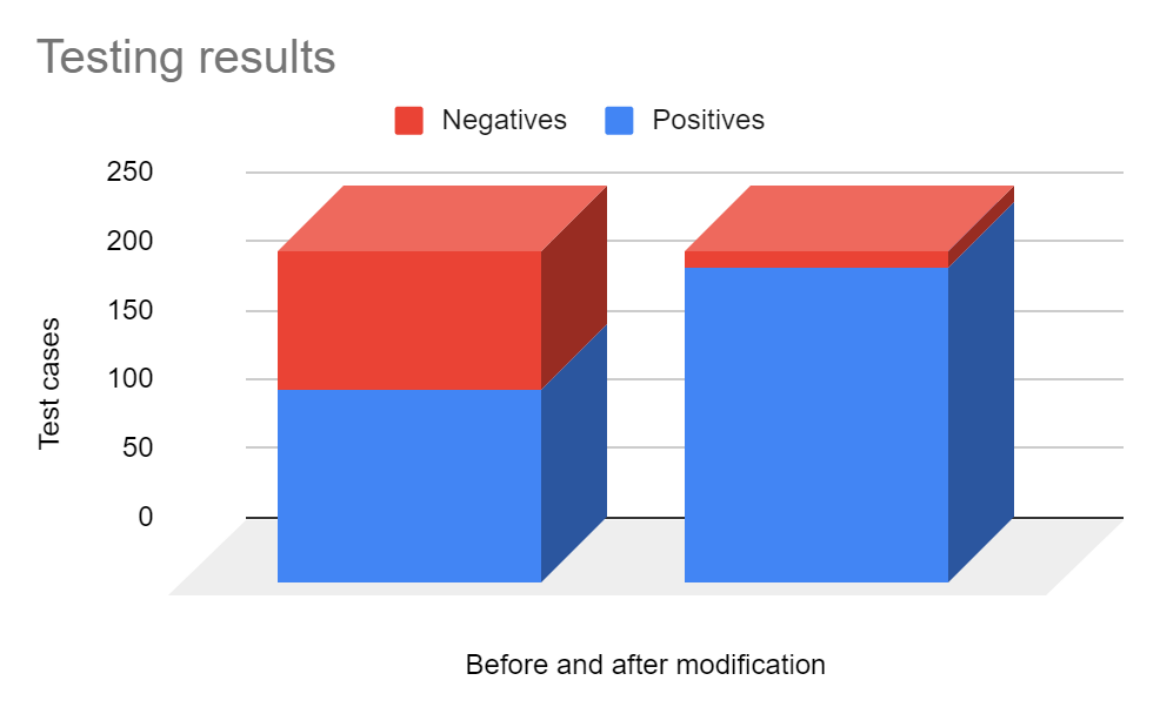
RESULTS

- Accuracy of NLP model:
  - The **F measure** is a **measure** of a test's accuracy and is defined as the weighted harmonic mean of the precision and recall of the test
  - Precision = TruePositives / (TruePositives + FalsePositives)
  - Recall = TruePositives / (TruePositives + FalseNegatives)
  - F-Measure = (2 * Precision * Recall) / (Precision + Recall)
  - Average F-measure value is 0.92445

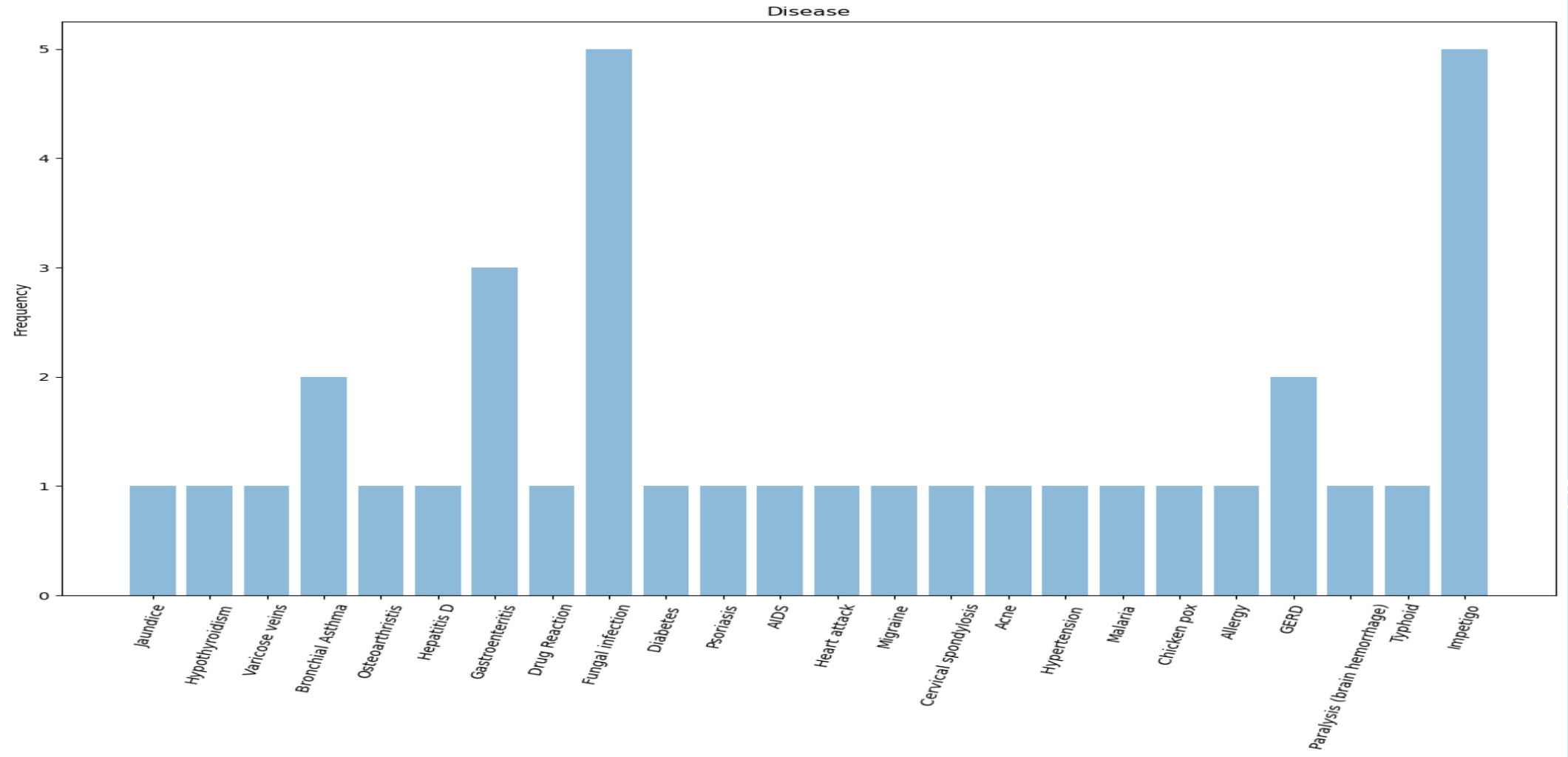


RESULTS

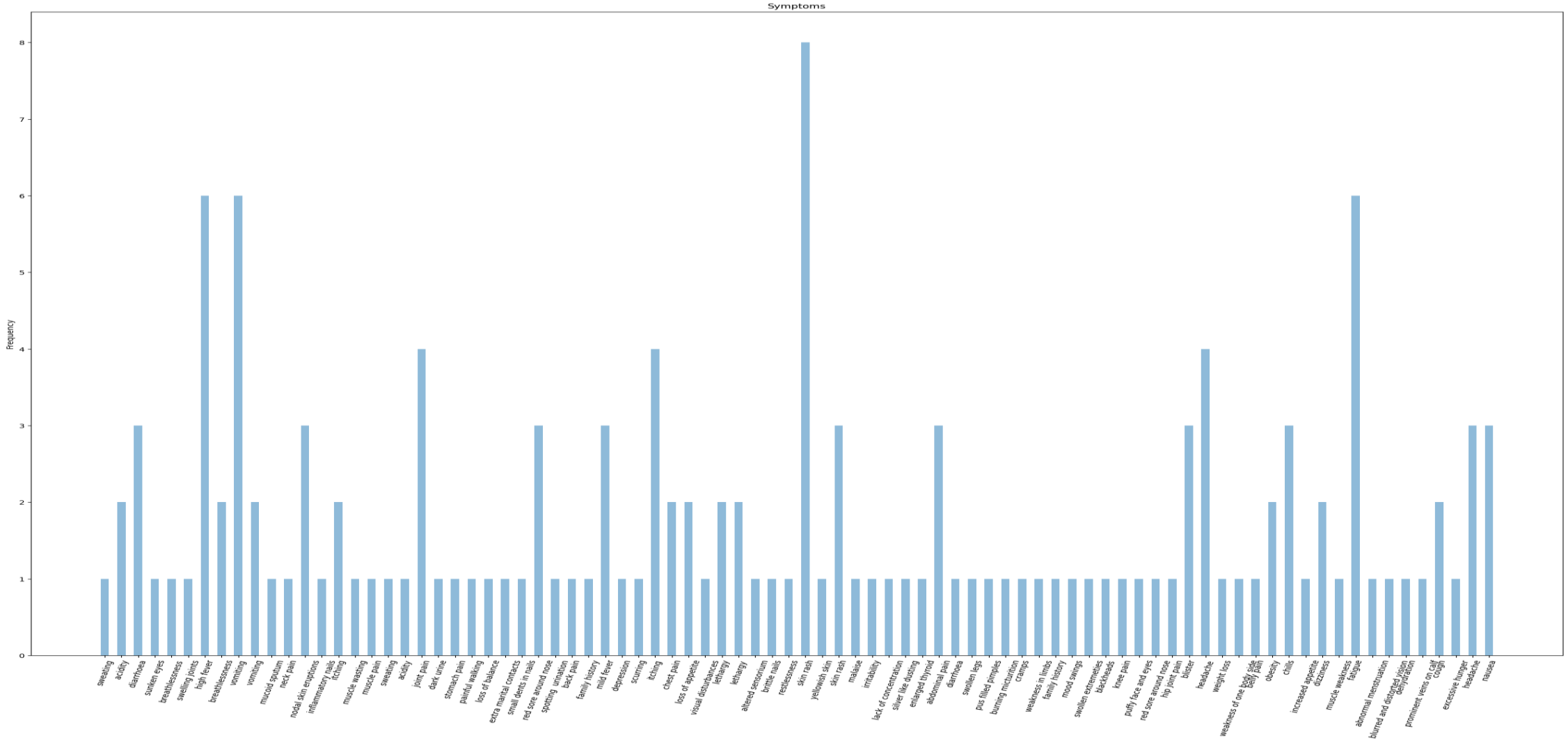
- Accuracy of classification part:
- The code was tested with several inputs and problems were noted down
- Initial accuracy was found to be 0.582 or 58.2%
- Corrections were made in the code to improve the results
- After modification, accuracy changed to 0.9502 or 95.02%



DATA ANALYSIS



DATA ANALYSIS



DATA ANALYSIS

```
In [14]: topdis=sorted(num_occ.items(), key=lambda x: x[1], reverse=True)
         topdis

Out[14]: [('Fungal infection', 5),
          ('Impetigo', 5),
          ('Gastroenteritis', 3),
          ('Bronchial Asthma', 2),
          ('GERD', 2),
          ('Jaundice', 1),
          ('Hypothyroidism', 1),
          ('Varicose veins', 1),
          ('Osteoarthristis', 1),
          ('Hepatitis D', 1),
          ('Drug Reaction', 1),
          ('Diabetes ', 1),
          ('Psoriasis', 1),
          ('AIDS', 1),
          ('Heart attack', 1),
          ('Migraine', 1),
          ('Cervical spondylosis', 1),
          ('Acne', 1),
          ('Hypertension ', 1),
          ('Malaria', 1),
          ('Chicken pox', 1),
          ('Allergy', 1),
          ('Paralysis (brain hemorrhage)', 1),
          ('Typhoid', 1)]
```

DATA ANALYSIS

```
Varicose veins
['prominent veins on calf', ' cramps', ' obesity', ' swollen legs']
count of symptom in disease:  Varicose veins  =  [(' cramps', 1)]
count of symptom in disease:  Varicose veins  =  [(' obesity', 2)]
count of symptom in disease:  Varicose veins  =  [(' swollen legs', 1)]
count of symptom in disease:  Varicose veins  =  [('prominent veins on calf', 1)]


Bronchial Asthma
['breathlessness', ' cough', ' mucoid sputum', ' family history', 'breathlessness', ' cough', ' mucoid sputum', ' family hist
ory', 'breathlessness', ' cough', ' mucoid sputum', ' family history']
count of symptom in disease:  Bronchial Asthma  =  [(' cough', 2)]
count of symptom in disease:  Bronchial Asthma  =  [(' mucoid sputum', 1)]
count of symptom in disease:  Bronchial Asthma  =  [('breathlessness', 1)]
count of symptom in disease:  Bronchial Asthma  =  [(' family history', 1)]


Osteoarthristis
['painful walking', ' swelling joints', ' hip joint pain', ' joint pain', ' knee pain']
```

PERT CHART

Aug 1st week
Forming the group

Aug 2nd week
Exploring and deciding the field of interest

Sept 1st week
Deciding the topic

Oct 1st week
Feasibility seminar

Oct 3rd week
Literature survey and confirming the dataset

Nov 2nd week
Phase 1 seminar

Nov 3rd week
Implementation

Jan last week
Integrating NLP, classification and UI

Feb 1st week
Hardware Implementation

March
Testing

April
Phase 2 seminar

May
Analysing data and modification

June
Completion of project and documentation

July
Final submission

CONCLUSION

- We intend to engineer an interactive health bot that listens to the symptoms and reciprocates with the disease and its' curative measures
- We also intend to store the data from this bot for carrying out further analytics on the health of the locality

REFERENCES

- [1] Chethana Saligram , Deepali B K , Gowri K S , J Ajay Kumar , Dr. Jyoti R Munavalli , "*Symptoms Extraction from a Voice Input using Natural Language Processing*", IJERT- Volume 09, Issue 04 (April 2020)
- [2]K. Jaiswal, S. Sobhanayak, B. K. Mohanta and D. Jena, "IoT-cloud based framework for patient's data collection in smart healthcare system using raspberry-pi," *2017 International Conference on Electrical and Computing Technologies and Applications (ICECTA)*, Ras Al Khaimah, 2017, pp. 1-4.
  doi: 10.1109/ICECTA.2017.8251967
- [3] Gustav Cederblad, "*Finding synonyms in medical texts – Creating a system for automatic synonym extraction from medical text"*, Linköping University, Bachelor thesis, 18 ECTS, Spring term 2018
- [4] Bhumika Gupta , *"Analysis of Various Decision Tree Algorithms for Classification in Data Mining" ,* International Journal of Computer Applications (0975 – 8887) Volume 163 – No 8, April 2017

THANK YOU !